

EXPLORATORY DATA ANALYSIS (EDA)

Objective: Extract insights using visual and statistical exploration.

Tools: Python(Pandas, Matplotlib, Seaborn)

1. Importing necessary libraries i.e, Pandas , Matplotlib, Seaborn & Loading the data using pandas.

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
data=pd.read_csv(r"C:\Users\Admin\Downloads\train.csv")
```

```
data
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

891 rows × 12 columns

2. Data Cleaning and Preprocessing

Performed using

- .describe()

- .info()
- .isnull().sum()
- .duplicated().sum()

➔ handled some missing values in Age, cabin, and Embarked column by using fillna() and drop() function.

➔ Converted Categorical columns(sex, Embarked).

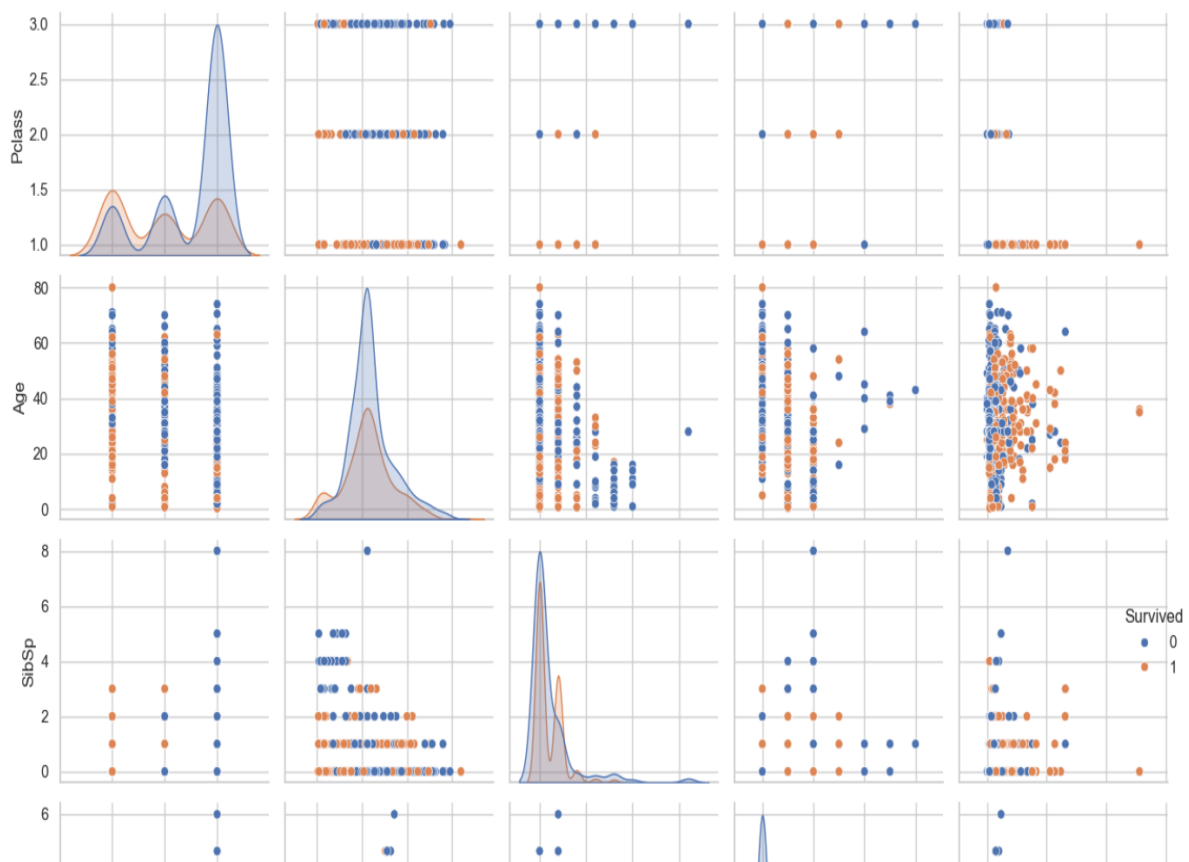
3. Pairplot for numeric features

```
sns.pairplot(data[['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare']],
hue='Survived')
```

```
plt.suptitle('Pairplot of Titanic Features', y=1.02)
```

```
plt.tight_layout()
```

```
plt.savefig('pairplot.png')
```



Findings for Pairplot:

Here we are comparing variables like Pclass, Age, SibSp, Fare, etc., and how they relate to Survived. The hue (Survived) is color-coded:

- **Orange** → Did not survive (0)
- **Blue** → Survived (1)

1. Pclass vs Survived

- There's a visible separation:
 - **Class 1 (left on x-axis)** has more blue → more survivors.
 - **Class 3 (right on x-axis)** has more orange → more deaths.
- **Higher class passengers had better survival chances.**

2. Age vs Survived

- Younger passengers (especially kids under 10) have more blue.
- Adults have mixed results, but older people (70+) seem to have mostly orange.
- **Younger passengers had a higher survival rate.**

3. Fare vs Survived

- Higher fares (towards right side of the Fare distribution) have a higher density of blue.
- **Passengers who paid more had better survival odds.**

4. SibSp (Siblings/Spouses Aboard)

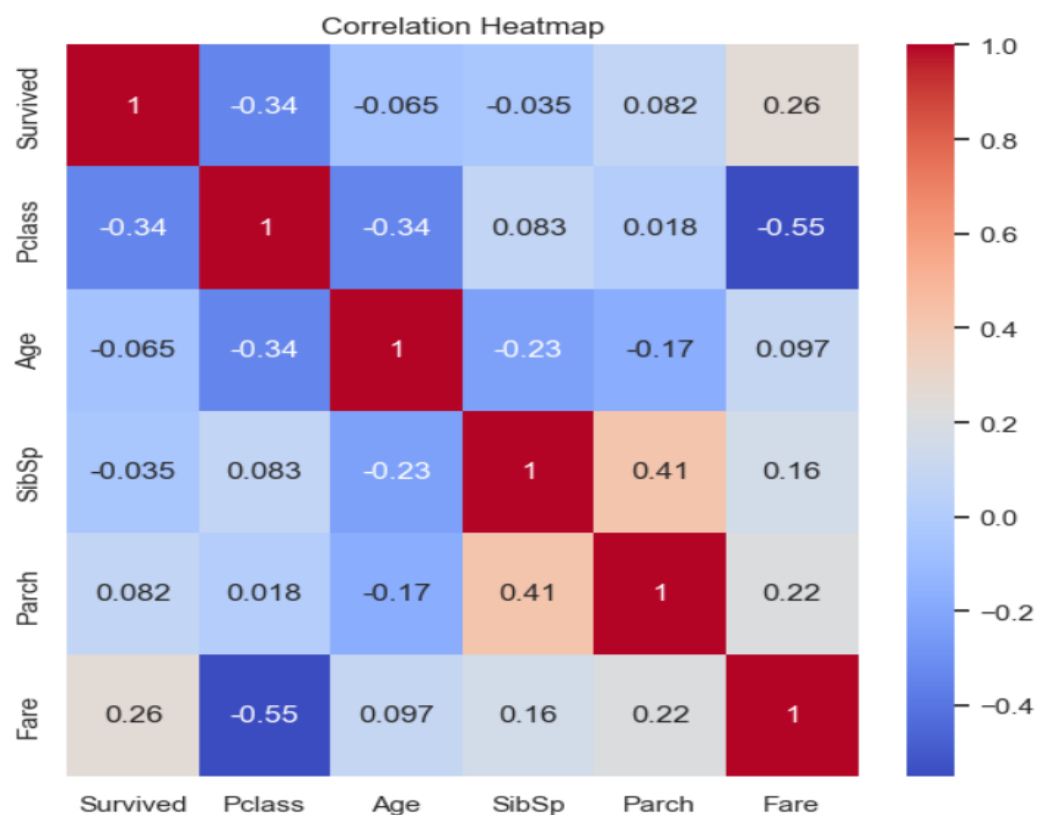
- Most people had 0 or 1 SibSp.
- There's a mild trend: having 1–2 family members may have helped; high numbers (like 5+) have low survival.
- **Being alone wasn't always worse, but big families didn't fare well.**

4. Heatmap

Findings of Heatmap:

The correlation heatmap provides insights into the linear relationships between different numerical features in the Titanic dataset. The strongest positive correlation with survival is **Fare (0.26)**, suggesting that passengers who paid higher fares were more likely to survive. There's a moderate **negative correlation between Pclass and Survived (-0.34)**, indicating that passengers in higher classes (lower Pclass number) had better survival chances. Other

variables like **Age** (-0.065) and **SibSp** (-0.035) have very weak negative correlations with survival, suggesting they had minimal linear influence individually. Additionally, **SibSp** and **Parch** (family-related features) show a positive correlation with each other (0.41), reflecting that passengers with siblings/spouses often had parents/children aboard as well. The **strongest overall correlation in the matrix** is between **Fare and Pclass** (-0.55), which makes sense as higher-class passengers typically paid more. Overall, this heatmap reinforces that **Pclass and Fare are among the most informative features for predicting survival**, while age and family size have weaker, possibly non-linear effects.

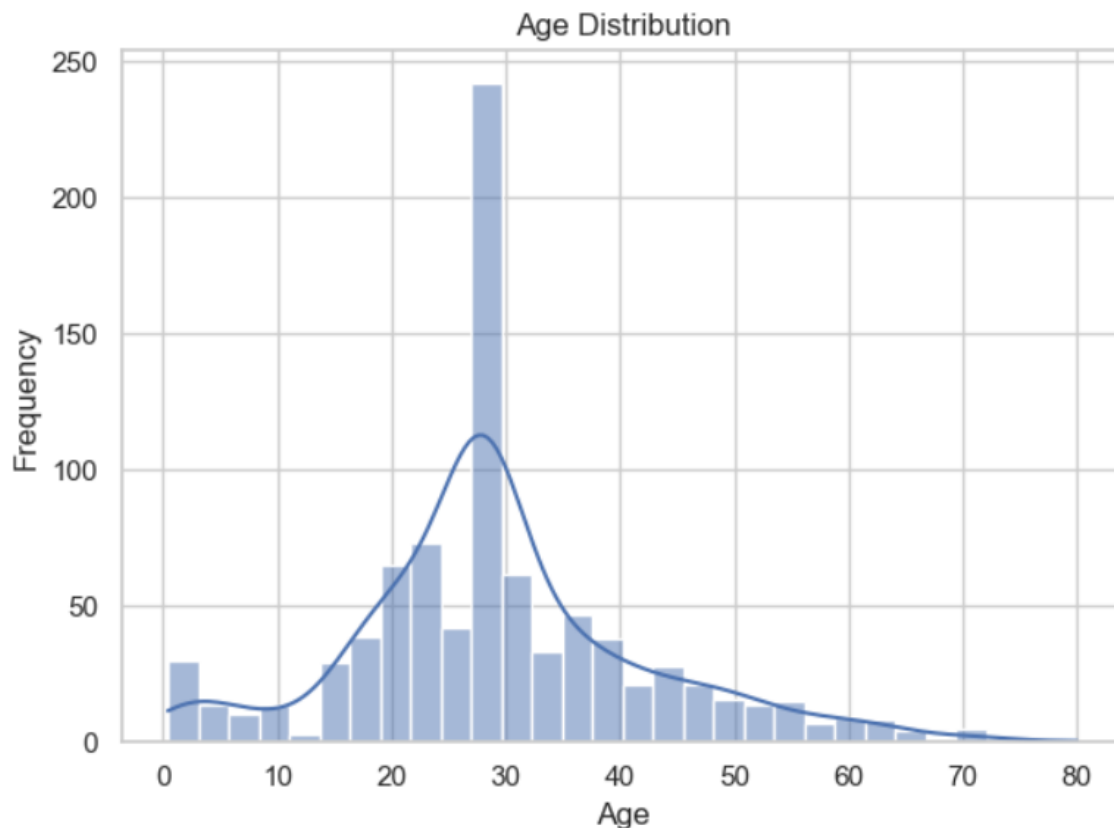


5. HISTOGRAM

Findings of Histogram:

The **Age Distribution** plot shows a right-skewed distribution, with the majority of passengers clustered between the ages of **20 and 40**, peaking sharply around **age 28–30**. This peak may indicate a high concentration of missing values that were imputed with the median age, which is common during data preprocessing. There's a noticeable decline in frequency beyond age 40, and very few passengers were above age 60. A small number of children under 10

were also present. Overall, the dataset is dominated by **young adults**, which is important to consider when analyzing survival patterns, especially since younger passengers might have had different survival rates compared to older adults or children.



6. BOXPLOT

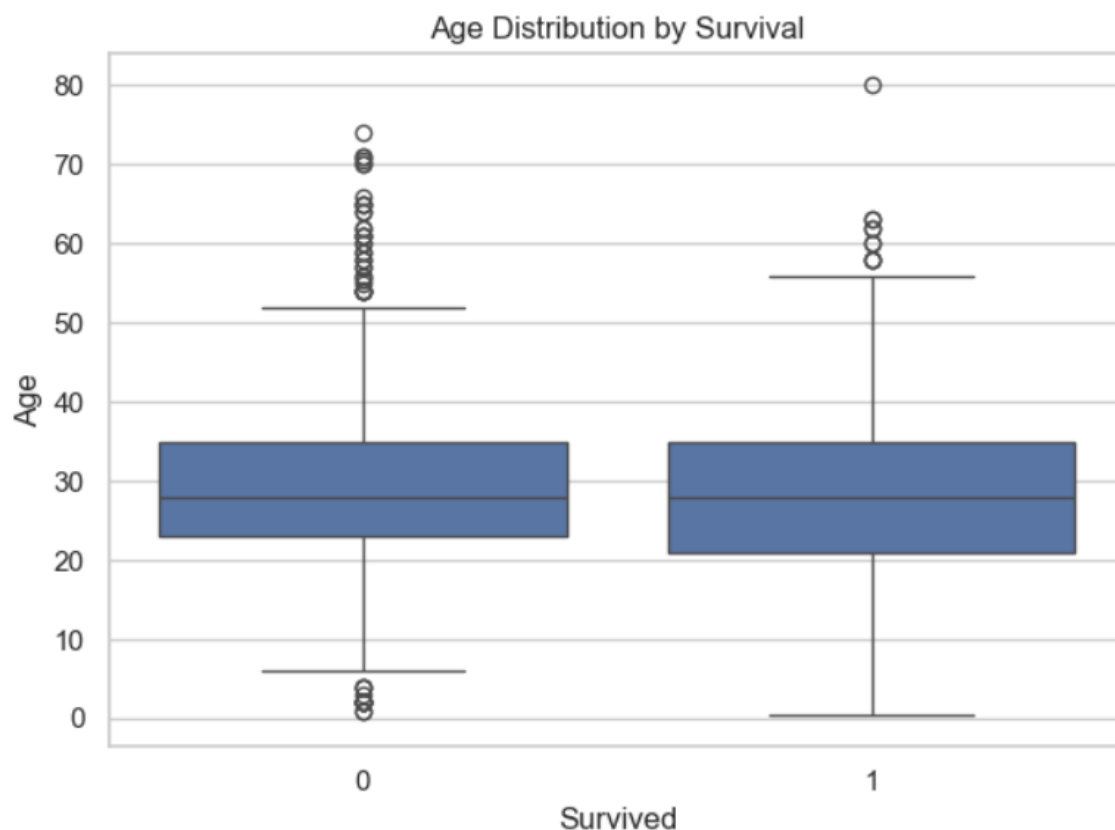
Findings of Boxplot:

The **boxplot of Age distribution by Survival** reveals subtle but meaningful differences between those who survived (1) and those who did not (0) on the Titanic:

- **Median Age** is slightly lower for survivors (~26) compared to non-survivors (~28).
- **Survivors show a wider spread of ages**, particularly toward younger passengers, including infants and children. In contrast, fewer very young passengers are seen among the non-survivors.
- **Outliers** exist in both groups, especially among the elderly (ages 60+), but survival among older individuals was relatively rare.

- The **interquartile ranges (IQRs)** for both groups are similar (roughly 20–38), suggesting that middle-aged adults made up a large portion of both survivors and non-survivors.

Although the difference in age between survivors and non-survivors isn't drastic, **younger passengers had slightly better survival odds**, and **very young children are more represented among survivors**, supporting the “women and children first” evacuation pattern observed in Titanic history.



7. SCATTERPLOT

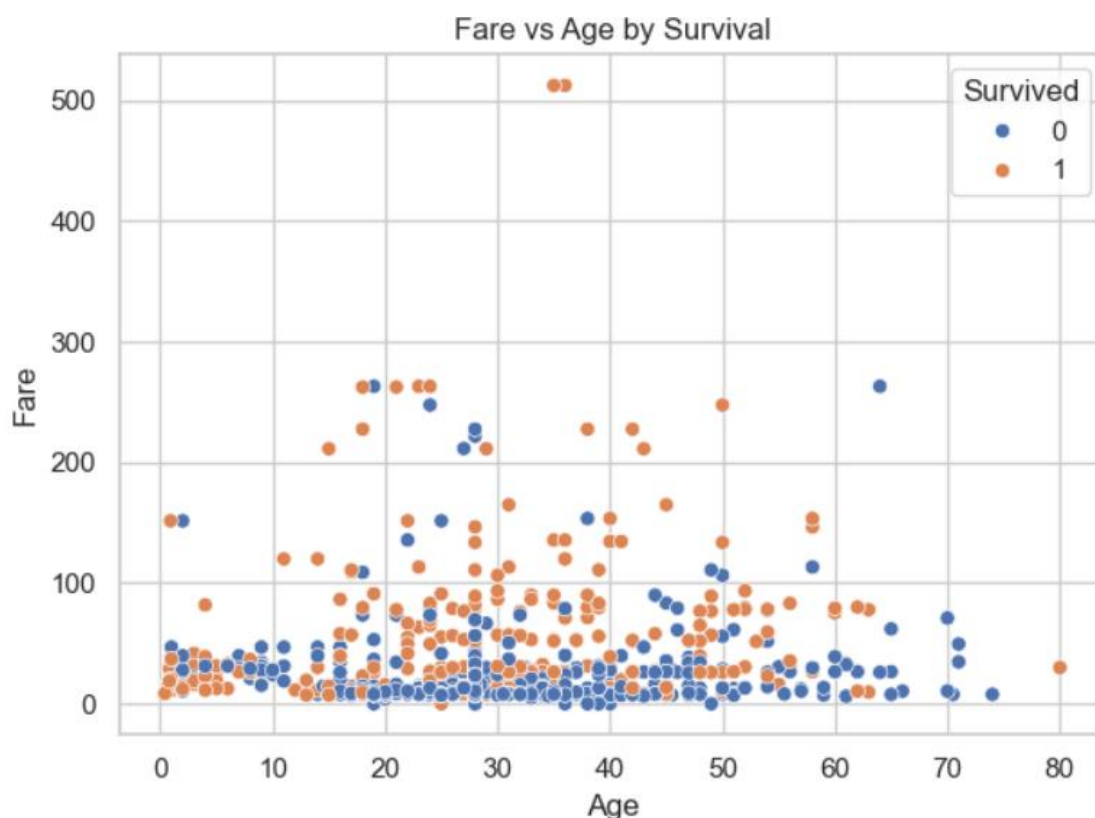
Findings of Scatterplot:

The **Fare vs. Age by Survival** scatter plot reveals interesting patterns about how age and fare relate to survival on the Titanic:

- **Survivors (orange dots)** are more concentrated in the **higher fare range**, especially for fares above **100**, indicating that passengers who paid more (likely in 1st class) had higher chances of survival.

- **Non-survivors (blue dots)** dominate the **lower fare range (0–50)**, suggesting that lower-class passengers (2nd or 3rd class) were less likely to survive.
- Across all ages, survivors are more widely distributed in terms of fare, while non-survivors are mostly clustered at low fare values.
- There are **some children and young adults (under age 20)** who paid high fares and survived, possibly from wealthy families.
- A few **elderly passengers** (above 60) are visible, but survival in this group was rare regardless of fare.

Higher fare generally correlates with better survival odds, likely due to better cabin locations and priority access to lifeboats. While age alone doesn't show a strong survival pattern, **fare is a more significant factor when predicting survival**.

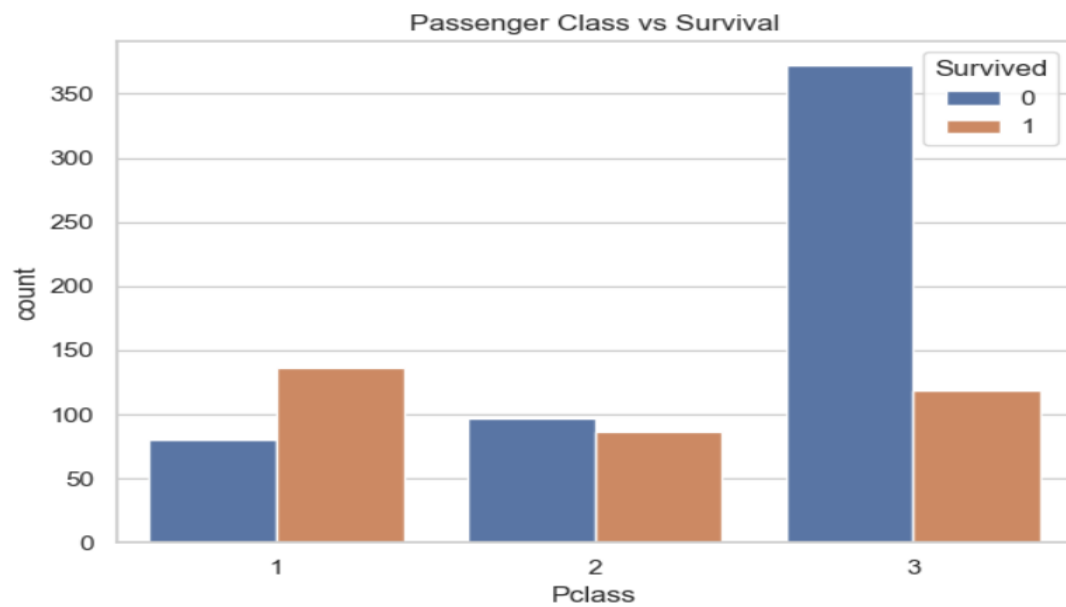


8. COUNTPLOT

Findings of Countplot:

1. Passenger Class vs Survival

- **Most 1st class** passengers survived, while **most 3rd class** passengers did not.
- Survival rates were highest in **Pclass 1**, followed by Pclass 2.
- **Pclass 3** had a significantly higher number of deaths compared to other classes.



2. Port of Embarkation vs Survival

- Most passengers embarked from **port S**, and a majority of them **did not survive**.
- Passengers from **port C** had a higher survival rate.
- The difference between survival and death is smaller in **port C**, suggesting better survival odds.

